

Introduction to Machine Learning

Homework 1

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Question 1

1.1

We substitute (24) into the objective function (23). For any training sample x_i , (24) gives

$$f(x) = \sum_{n=1}^N \alpha_n K(x, x_n),$$

where $K_{i,n} = K(x_i, x_n)$. Therefore, the vector of predictions on all training inputs can be written as

$$\begin{bmatrix} f(x_1) \\ f(x_2) \\ \vdots \\ f(x_N) \end{bmatrix} = \begin{bmatrix} \sum_{n=1}^N \alpha_n K(x_1, x_n) \\ \sum_{n=1}^N \alpha_n K(x_2, x_n) \\ \vdots \\ \sum_{n=1}^N \alpha_n K(x_N, x_n) \end{bmatrix} = K\alpha.$$

Hence, the empirical risk (sum of squared errors) becomes

$$\sum_{n=1}^N (y_n - f(x_n))^2 = \|y - K\alpha\|_2^2, \text{ where } y = [y_1, \dots, y_N]^\top$$

Next, to compute the regularization term $\|f\|_{\mathcal{H}}^2$, we use the RKHS inner-product property. Since $f = \sum_{n=1}^N \alpha_n K(\cdot, x_n)$, by linearity,

$$\|f\|_{\mathcal{H}}^2 = \langle f, f \rangle_{\mathcal{H}} = \left\langle \sum_{n=1}^N \alpha_n K(\cdot, x_n), \sum_{m=1}^N \alpha_m K(\cdot, x_m) \right\rangle_{\mathcal{H}} = \sum_{n=1}^N \sum_{m=1}^N \alpha_n \alpha_m \langle K(\cdot, x_n), K(\cdot, x_m) \rangle_{\mathcal{H}}.$$

By the reproducing property, $\langle K(\cdot, x_n), K(\cdot, x_m) \rangle_{\mathcal{H}} = K(x_n, x_m)$, and thus

$$\|f\|_{\mathcal{H}}^2 = \sum_{n=1}^N \sum_{m=1}^N \alpha_n \alpha_m K(x_n, x_m) = \alpha^\top K \alpha.$$

Therefore, the objective (23) is transformed into

$$\min_{\alpha \in \mathbb{R}^N} \|y - K\alpha\|_2^2 + \lambda \alpha^\top K \alpha,$$

which is exactly (25).

1.2

We now derive the chain rule for $\nabla_h L$ when using the RBF kernel

$$K(x_i, x_j) = \exp\left(-\frac{\|x_i - x_j\|_2^2}{h}\right), \quad K \in \mathbb{R}^{N \times N}, \quad K_{ij} = K(x_i, x_j).$$

Therefore,

$$\frac{\partial K_{ij}}{\partial h} = \frac{\|x_i - x_j\|_2^2}{h^2} K_{ij}.$$

Define $D \in \mathbb{R}^{N \times N}$ by $D_{ij} = \|x_i - x_j\|_2^2$ and let $\odot$ denote the Hadamard product. Then in matrix form

$$K' \triangleq \frac{\partial K}{\partial h} = \frac{1}{h^2} (D \odot K).$$

We know that

$$L(\alpha, h) = \|y - K(h)\alpha\|_2^2 + \lambda \alpha^\top K(h)\alpha.$$

Let $r = y - K\alpha$. Then

$$L = r^\top r + \lambda \alpha^\top K \alpha, \quad \frac{\partial r}{\partial h} = -K' \alpha.$$

Therefore,

$$\nabla_h L = \frac{\partial L}{\partial h} = 2r^\top \frac{\partial r}{\partial h} + \lambda \alpha^\top K' \alpha = -2(y - K\alpha)^\top K' \alpha + \lambda \alpha^\top K' \alpha,$$

where $K' = \frac{1}{h^2} (D \odot K)$, $D_{ij} = \|x_i - x_j\|_2^2$.

Question 2

To encourage a sparse coefficient vector α , I use the ℓ_1 regularizer

$$\mathcal{R}(\alpha) = 2\|\alpha\|_1 = 2 \sum_{i=1}^N |\alpha_i|,$$

which leads to the sparse kernel regression problem

$$\min_{\frac{1}{2}\alpha \in \mathbb{R}^N} \|y - K\alpha\|_2^2 + \lambda \|\alpha\|_1.$$

Then, I use coordinate descent with iterative soft-thresholding. Let $K_i \in \mathbb{R}^N$ denote the i -th column of K , and let K_{-i} be the matrix K with column i removed (similarly α_{-i} is α with entry i removed). Define the soft-thresholding operator

$$\text{SoftThreshold}_\tau(z) = \text{sign}(z) \max\{|z| - \tau, 0\}.$$

Iterative soft-thresholding:

- **Initialization:** start with $\alpha^{(0)} = 0$.
- **Iteration:** for $t = 0, 1, 2, \dots$, sweep $i = 1, \dots, N$ and update α_i while keeping the others fixed. Let the partial residual be

$$r_{-i}^{(t)} = y - K_{-i}\alpha_{-i}^{(t)}.$$

Then update

$$\alpha_i^{(t+1)} = \text{SoftThreshold}_{\lambda/(\|K_i\|_2^2)} \left(\frac{K_i^\top r_{-i}^{(t)}}{\|K_i\|_2^2} \right) = \text{SoftThreshold}_{\lambda/(\|K_i\|_2^2)} \left(\frac{K_i^\top (y - K_{-i}\alpha_{-i}^{(t)})}{\|K_i\|_2^2} \right).$$

- **Convergence:** repeat the sweeps until convergence.